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1. Device and the serial interface commands

S.N.	Command	Value	Reboot	Purpose
1	Reset	0	+⚠️	Restart or Factory Reset EzloPi Device
2	Get Info	1	-	Get EzloPi Device Info
3	Set WiFi	2	-	Set/Configure WiFi

4	Set Config	3	-	Set/Configure Devices and Sensors based on interface
5	Get Config	4	-	Request for stored configuration about Sensors and Devices for different interfaces
6	Set Serial	5	+⚠️	Set/Configure Serial Port protocol details

a. ## Command : Reset

This command is for device administrative purpose such as device restart, device factory restart etc

i. ### Request from UI

cmd-0, request from UI

```
{
    "cmd": 0,
    "sub_cmd": 0
}
```

ii. ### Response from EzloP Device

cmd-0, response from UI

```
{
  "cmd": 0,
  "status": 1
}
```

iii. ### Key – Value description

S.N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	0: Reset 1: Info, 2: Set WiFi
2	sub_cmd	Sub command to main command	Number	0: Factory Restore 1: Reboot Device
Response From Device				
1	cmd	Command Code/ID	Number	0: Restart 1: Info, 2: Set WiFi
2	status	Status of the request	Number	0 : Failed issuing command, 1 : Success

b. ## Command: Get Info

i. ### Request from UI

cmd-1 request from UI

```
{
"cmd": 1
}
```

ii. ### Response from EzloP Device

cmd-1 response from UI


```
{
    "cmd": 1,
    "api": "1.0.0",
    "info": {
        "ezlopi_firmware": {
            "version": "3.3.6",
            "build": 185,
            "build_date": "2024-02-29T18:46:33+545"
        },
        "chip": {
            "type": "ESP32",
            "version": "1.00"
        },
        "firmware_sdk": {
            "name": "ESP-IDF",
            "version": "v4.4.6-dirty"
        },
        "device_state": {
            "uptime": "0d 2h 55m 9s",
            "boot_count": 1,
            "boot_reason": "Power ON",
            "flash_size": "4MB",
            "wifi_mac": "EC:62:60:A7:1D:98",
            "ble_mac": "EC:62:60:A7:1D:9A"
        },
        "serial_config": {
            "baud": 115200,
            "parity": "N",
            "start_bits": 0,
            "stop_bits": 1,
            "frame_size": 8,
            "flow_control": "DISABLED"
        },
        "ezlopi_cloud": {
            "provisioned": true,
            "uuid": "1b576400-8392-11ee-8889-0f56091e7639",
            "uuid_prov": "c2ea484b-6b31-446e-b42e-8ab7b5c36ce4",
            "type": "ezlopi_generic",
            "serial": 102002167,
            "mac": "a2:f8:75:1c:3a:4c"
        },
        "oem": {
            "brand": "",
            "manufacturer": "",
            "model": ""
        },
        "network": {
            "ssid": "nepaldigisys",
            "wifi_mode": "STA",
            "ip_sta": "192.168.1.96",
            "ip_nmask": "255.255.255.0",
            "ip_gw": "192.168.1.254",
            "wifi": true,
            "internet": true,
            "cloud": true
        }
    }
}
```


i. Key – Value description

S. N.	Key	Details	Type	*Mandatory	Example values	Possible values

1	cmd	Command Code/ID	Number	+	1: Info, 2: Set WiFi	
2	api	Serial API Version	String	-	1.0.0	
3	info	Device Detailed Information	JSON Object	+		
4	info.ezlopi_firmware	EzloPi Firmware Info	JSON Object	+		
5	info.ezlopi_firmware.version	EzloPi Firmware Version	String	+	3.4.6	
6	info.ezlopi_firmware.build	EzloPi Firmware Build Number	Number	+	186	
7	info.ezlopi_firmware.build_date	EzloPi Firmware Build Date (ISO 8601: YYYY-MM-DDThh:mm:ss±hhmm)	String	+	2024-02-29T18:46:33+545	
8	info.chip	Microcontroller (Chip) Info	JSON Object	-		
9	info.chip.type	Microcontroller (Chip) type	String	-	ESP32	
10	info.chip.version	Microcontroller (Chip) Architecure version provided by manufacturer	String	-	1.00	
11	info.firmware_sdk	Microcontroller HAL SDK/API info provided by the manufacturer	JSON Object	-		
12	info.firmware_sdk.name	API /SDK Name	String	-	ESP-IDF	
13	info.firmware_sdk.version	API /SDK Version	String	-	v4.4.6-dirty	
14	info.device_state	EzloPi Device Current Status	JSON Object	+		
15	info.device_state.uptime	Time since device is ON	String	+	0d 2h 55m 9s	
16	info.device_state.boot_count	Number of times the device has been rebooted after firmware has been flashed into it.	Number	-	7	
17	info.device_state.boot_reason	Cause of last boot	String	-	Power ON	
18	info.device_state.flash_size	Total flash memory on the device	String	-	4MB	
19	info.device_state.wifi_mac	EzloPi Device WiFi MAC Address	String	+	EC:62:60:A7:1D:98	
20	info.device_state.ble_mac	EzloPi Device Bluetooth MAC Address	String	+	EC:62:60:A7:1D:9A	
21	info.serial_config	Serial Port Configuration	JSON Object	+		
22	info.serial_config.baud	Currently configured serial port baud rate	Number	+	115200	
23	info.serial_config.parity	Currently configured serial port parity bit	String	+	N	1. N : No Parity 2. P : Parity Odd 3. E : Parity Even
24	info.serial_config.start_bits	Currently configured serial port start bits	Number	+	0	Default 0

25	info.serial_config.stop_bits	Currently configured serial port stop bits	Number	+	1	Ranges in between 1 - 2. Default 1
26	info.serial_config.frame_size	Currently configured serial port frame size	Number	+	8	Frame size ranges from 5 bits - 8 bits, Default 8.
27	info.serial_config.flow_control	Currently configured serial port flow control	String	-	DISABLED	1. DISABLED : Hardware Flow Control Disabled 2. RTS : Hardware Flow Control RTS 3. CTS : Hardware Flow Control CTS 4. CTS_RTS : Hardware Flow Control CTS and RTS Flow control  Hardware flow control is not yet supported on EzloPi, so the value needs to be "DISABLED".
28	info.ezlopi_cloud	Ezlo Cloud related info for EzloPi	JSON Object	+		
29	info.ezlopi_cloud.provisioned	Provisioned Status	Boolean	+	True	
30	info.ezlopi_cloud.uuid	Device UUID created on Ezlo Cloud	String	+	1b576400-8392-11ee-8889-0f56091e7639	
31	info.ezlopi_cloud.uuid_prov	Device provisioning UUID created on Ezlo Cloud	String	+	c2ea484b-6b31-446e-b42e-8ab7b5c36ce4	
32	info.ezlopi_cloud.type	EzloPi Device Type	String	+	ezlopi_generic	Default : ezlopi_generic
33	info.ezlopi_cloud.serial	Ezlo Cloud device ID or Device Serial Number	Number	+	102002167	Range : 102000000 - 102999999
34	info.ezlopi_cloud.mac	MAC created for Ezlo Device on Cloud	String	+	a2:f8:75:1c:3a:4c	
35	info.oem	OEM related info	JSON Object	-		
36	info.oem.brand	Brand Name	String	-		OEM Brand Name  Maximum 64 Characters
37	info.oem.manufacturer	Manufacturer Name	String	-		OEM Manufacturer name  Maximum 64 Characters
38	info.oem.model	Model Number	String	-		OEM Model Name  Maximum 64 Characters
39	info.network	Current network related info on EzloPi	JSON Object	+		
40	info.network.ssid	WiFi SSID (WiFi must be 2.4GHz)	String	+	myWiFiSSID	WiFi SSID Max Char  Maximum 32 Characters
41	info.network.wifi_mode	WiFi Password	String	+		WiFi Password Max Char  Maximum 64 Characters
42	info.network.ip_sta	WiFi (LAN) IP obtained from DHCP Server	String	+	192.168.1.96	
43	info.network.ip_nmask	LAN Netmask IP	String	-	255.255.255.0	

44	info.network.ip_gw	LAN Gateway IP	String	-	192.168.1.254	
45	info.network.wifi	WiFi connection status	Boolean	+	true	
46	info.network.internet	Internet connection status	Boolean	+	true	
47	info.network.cloud	Ezlo cloud connection status	Boolean	+	true	

Mandatory Note

Mandatory signifies that firmware is supposed to return all the mandatory values (indicated with +) and can ignore optional (indicated with -).

c. Command: Set Wi-Fi

i. Request from UI

cmd-2 request from UI

```
{
  "cmd": 2,
  "ssid": "my wifi ssid",
  "pass": "myWiFiPass"
}
```

ii. Response from EzloP Device

cmd-1 response from device

```
{
  "cmd": 2,
  "status_write": 1,
  "status_connect": 1
}
```

iii. Key – Value description

S.N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device
2	ssid	WiFi SSID	String	My Example SSID
3	pass	WiFi Password for SSID	String	MyWiFi1Pass
Response From Device				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device
2	status_write	Write Status on device non-volatile memory (flash)	Number	0: Failed, 1: Success
3	status_connect	WiFi connection Status with any of two of the network	Number	0: Failed, 1: Success with SSID1, 2: Success with SSID 2

d. Command: Set Config

Config changes on 3.5.8 B 187

Config JSON has been changed :
+ "chipset": "ESP32"
- "dev_total": 3,

i. **Request from UI**

cmd-3 request from UI

```
{
    "cmd": 3,
    "config_id": "_ID_",
    "config_time": 1696508363,
    "config_name": "My moisture sensor",
    "chipset": "ESP32",   // ESP32, ESP32S3 or ESP32C3
    "dev_detail": [
        {
            "dev_type": 1,
            "dev_name": "Dining Room Main Lamp",
            "id_room": "",
            "id_item": 2,
            "val_ip": true,
            "val_op": false,
            "gpio_in": 21,
            "gpio_out": 13,
            "is_ip": false,
            "ip_inv": true,
            "pullup_ip": true,
            "pullup_op ": true,
            "op_inv": false
        },
        {
            "dev_type": 2,
            "dev_name": "Switch",
            "id_room": "",
            "id_item": 4,
            "gpio": 21,
            "pull_up": true,
            "logic_inv": true
        },
        {
            "dev_type": 8,
            "dev_name": "Sensor for my Dog",
            "id_room": "",
            "id_item": 5,
            "gpio_sda": 21,
            "gpio_scl": 22,
            "pullup_scl": true,
            "pullup_sda": true,
            "slave_addr": 105
        }
    ]
}
```

ii. **Response from EzloP Device**

cmd-3 response from device

```
{
  "cmd": 3,
  "status_write": 1
}
```

iii. **Key – Value description**

S.N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	0: Restart, 1: Info, 2: Set WiFi, 3: Add Device, 4: Get Config

2	chipset	Compatible chip for the configuration	String	ESP32, ESP32S3 or ESP32C3
3	config_id	Any random number within this range is fine.	Number	1098907645
4	config_time	Epoch time while flashing	Number	1696508363
5	config_name	Name of this configuration in a sense user can save this config or sometime re-use config he/she shaved earlier	String	My moisture sensor
6	dev_details	Array of devices with all the defined properties	JSON Object	
Response From EzloPi Device				
1	cmd	Command Code/ID	Number	0: Reset, 1: Info, 2: Set WiFi, 3: Add Device, 4: Get Config
2	status	Config Write Status	Number	0: Failed, 1 Success

e. ## Command : Get Config

- ### Request from UI

cmd-4 request from UI

```
{
  "cmd": 4
}
```

- ### Response from EzloP Device

Config changes on 3.5.8 B 187



Config JSON has been changed :
+ "chipset": "ESP32"
- "dev_total": 3,

cmd-4 response from device


```
{
    "cmd": 4,
    "config_id": "_ID_",
    "config_time": 1696508363,
    "config_name": "My moisture sensor",
    "chipset": "ESP32",
    "dev_detail": [
        {
            "dev_type": 1,
            "dev_name": "Dining Room Main Lamp",
            "id_room": "",
            "id_item": 2,
            "val_ip": true,
            "val_op": false,
            "gpio_in": 21,
            "gpio_out": 13,
            "is_ip": false,
            "ip_inv": true,
            "pullup_ip": true,
            "pullup_op ": true,
            "op_inv": false
        },
        {
            "dev_type": 2,
            "dev_name": "Switch",
            "id_room": "",
            "id_item": 4,
            "gpio": 21,
            "pull_up": true,
            "logic_inv": true
        },
        {
            "dev_type": 8,
            "dev_name": "Sensor for my Dog",
            "id_room": "",
            "id_item": 5,
            "gpio_sda": 21,
            "gpio_scl": 22,
            "pullup_scl": true,
            "pullup_sda": true,
            "slave_addr": 105
        }
    ]
}
```


- **Key – Value description**

S.N.	Key	Details	Type	Data
		Request From UI		
1	cmd	Command Code/ID	Number	0: Reset, 1: Info, 2: Set WiFi, 3: Add Device, 4: Get Config
		Response From Device		
1	cmd	Command Code/ID	Number	0: Reset, 1: Info, 2: Set WiFi, 3: Add Device, 4: Get Config
2	status	Read status from device's non-volatile memory (flash)	Number	0: Failed, 1 Success
3	dev_detail	Array of devices with all the defined properties	JSON Object	
4	config_id	Any random number within this range is fine.	Number	1098907645
5	chipset	Compatible chip for the configuration	String	ESP32, ESP32S3 or ESP32C3

6	config_time	Epoch time while flashing	Number	1696508363
7	config_name	Name of this configuration in a sence user can save this config or sometime re-use config he/she shaved earlier	String	My moisture sensor

f. ## Command: Set UART Configuration

i. ### Request from UI

cmd-2 request from UI

```
{
    "cmd": 5,
    "baud": 115200,
    "parity": "N",
    "start_bits": 0,
    "stop_bits": 1,
    "frame_size": 8,
    "flow_control": "DISABLED"
}
```

ii. ### Response from EzloP Device

cmd-1 response from device

```
{
  "cmd": 5,
  "status": 1
}
```

iii. ### Key – Value description

S. N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device
2	baud	Baud Rate to be set	Number	
3	parity	Parity to be set	String	1. N : No Parity 2. P : Parity Odd 3. E : Parity Even
4	start_bits	Number of bits before starting the data bits	Number	Default 0
5	stop_bits	Number of bits after data bits	Number	Ranges in between 1 - 2. Default 1
6	frame_size	Total number of data bits in single frame	Number	Frame size ranges from 5 bits - 8 bits, Default 8.
7	flow_control	Hardware Flow control	String	1. DISABLED : Hardware Flow Control Disabled 2. RTS : Hardware Flow Control RTS 3. CTS : Hardware Flow Control CTS 4. CTS_RTS : Hardware Flow Control CTS and RTS Flow control Hardware flow control is not yet supported on EzloPi, so the value needs to be "DISABLED".
Response From Device				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device
2	status	Command execution status	Number	0: Failed, 1: Success

g. ## Command: Log Configuration

Command Log History

This feature has been added after the version 3.5.6 Build 187.

i. Request from UI

Log configuration

```
{
    "cmd": 6,
    "target": 1,                // 0 = Cloud , 1 = Serial
    "severity": 1               // 0 = NONE, 1= ERROR,  2 = WARNING, 3 = INFO, 4 = DEBUG,
5 = TRACE
}
```

ii. Response from EzloPi Device

Response : Log Command

```
{
  "cmd": 6,
  "status": 1         // 0 = failed, 1 = success
}
```

iii. Key-Value Description

S. N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device, ... 6. Log
2	target	Log target, either cloud or Serial (Local)	Number	0 = Cloud , 1 = Serial
3	severity	Log Severity level	Number	0 = NONE, 1= ERROR, 2 = WARNING, 3 = INFO, 4 = DEBUG, 5 = TRACE
Response From Device				
1	cmd	Command Code/ID	Number	1: Info, 2: Set WiFi, 3: Add Device, ... 6. Log
2	status	Command execution status	Number	0: Failed, 1: Success

2. EzloPi Device and Sensors Interface Details

a. Interfaces Categories with their ID

S.N.	dev_type	Interface ID
1	Digital Output	1
2	Digital Input	2
3	Analog Input (ADC)	3
4	Analog Output (DAC)	4
5	PWM	5
6	UART	6
7	One Wire	7
8	I2C	8
9	SPI	9
10	Other	10

b. ## Devices and Sensors supported with their ID

List of Sensors and Devices Support available on EzloPi

S.N.	id_item	Device / Sensor ID	Interface	Interface ID
1	LED	1	Digital Out	1
2	Relay	2	Digital Out	1
3	Plug	3	Digital Out	1
4	Switch / Push Button / Key	4	Digital In	2
5	MPU6050 Accelerometer and Gyroscope	5	I2C	8
6	ADXL345 3-Axis Accelerometer	6	I2C	8
7	GY-271 Magnetometer	7	I2C	8
8	LTR303ALS Luminosity sensor	8	I2C	8
9	RMT SK6812	9	Other	
10	BMP680 I2C Sensor	10	I2C	8
11	BMP280 I2C Sensor	12	I2C	8
12	DHT11 Temperature Humidity Sensor	15	One Wire	7
13	DHT22 Temperature Humidity Sensor	16	One Wire	7
14	Potentiometer	17	Analog to Digital	3
15	Internal Hall Effect Sensor	18	Other	10
16	PIR Sensor	19	Digital In	2
17	Parallax 2-Axis Joystick	20	Other	10
18	MB1013 Sonar	21	UART	6
19	PWM Generic Load	22	PWM	5
20	Touch Switch TTP223B (Toggle)	23	Digital In	2
21	Ultrasonic Sensor HC SR04	24	Other	10
22	LDR Digtial Sensor	25	Digital In	2
23	LDR analog Sensor	26	Analog to Digital	3
24	Water Leak Sensor	27	Analog to Digital	3
25	GY61 3-Axis Accelerometer	28	Other	10
26	GXHTC3 - RH and T Sensor	29	I2C	8
27	DS18B20 Temperature Sensor	30	One Wire	7
28	JSN SR04T Water Level Sensor	31	Other	10
29	Soil Moisture Sensor	32	Analog to Digital	3
30	Turbidity Sensor	33	Analog to Digital	3
31	Proximity Sensor	34	Digital In	2
32	Touch switch TPP 223B (Momentary)	35	Digital In	2
33	Servo Motor MG996R	36	PWM	5

34	SK6812 RGB LEDs	38	Other	10
35	TCS230 Sensor	40	PWM	5
36	FC28 Soil Moisture Sensor	41	Analog to Digital	3
37	Shunt voltmeter	42	Analog to Digital	3
38	UV Intensity Sensor	43	Analog to Digital	3
39	TSL2561 Luminosity Sensor	44	I2C	8
40	CS712TELC 05B Current Sensor	46	Analog to Digital	3
41	HX711 LOADCELL	47	Other	10
42	MQ4 Methane Gas Sensor	48	Other	10
43	MQ2 LPG Gas Sensor	49	Other	10
44	MQ3 Alcohol Sensor	50	Other	10
45	MQ8 H2 Gas Sensor	51	Other	10
46	MQ135 NH3 Gas Sensor	52	Other	10
47	GPS GYGPS6MV2	53	UART	6
48	YFS201 flowmeter	54	PWM	5
49	Flex Resistor	55	Analog to Digital	3
50	Force Sensitive Resistor	56	Analog to Digital	3
51	KY026 Flame Sensor	57	Other	10
52	MQ6 LPG Detector	59	Other	10
53	Vibration Sensor	60	Digital In	2
54	Reed Switch	61	Digital In	2
55	MQ7 CO Detector	62	Other	10
56	MQ9 LPG Detector	63	Other	10
57	Float Switch	65	Digital In	2
58	R307 Fingerprint Module	66	Other	10

3. Configuration Details for each of the device (interface) type
    a. **Digital Output**
        i. **JSON Structure**

Digital Out JSON

```
{
 "dev_type": 1,
  "dev_name": "Dining Room Main Lamp",
  "id_room":"",
  "id_item": 2,
  "val_ip": true,
  "val_op": false,
  "gpio_in": 21,
  "gpio_out": 13,
  "is_ip": false,
  "ip_inv": true,
  "pullup_ip": true,
  "pullup_op ": true,
  "op_inv": false
}
```

ii. Key Value description

S. N.	Key	Details	Type	Data
1	val_ip	Default Input Value	Boolean	True : HIGH, False : LOW
2	val_op	Default Output Value	Boolean	True : HIGH, False : LOW
3	gpio_in	GPIO Input pin number	Number	15
4	gpio_out	GPIO Output pin number	Number	12
5	is_ip	Is Input enabled	Boolean	True : Input has been enabled, use the pin number as specified in the key gpio_in
6	ip_inv	Is Input in Inverted logic	Boolean	True : Output operates on inverted logic, False : Normal
7	op_inv	Is output in Inverted logic	Boolean	True : input operates on inverted logic, False : Normal

b. Digital Input:

i. JSON Structure

Digital In JSON

```
{
        "dev_type": 2,
        "dev_name": "Switch",
        "id_room":"",
        "id_item": 4,
        "gpio": 21,
        "pull_up": true,
        "logic_inv": true
}
```

ii. Key Value description

S.N.	Key	Details	Type	Data
1	gpio	GPIO Input pin number	Number	15
2	pull_up	Enable Pull up or Enable Pull Down	Boolean	True : Enable Pull-Up, False : Enable Pull Down
3	logic_inv	Is switch operates in Inverted logic	Boolean	True : input operates on inverted logic, False : Normal

c. Analog Input (ADC)

i. JSON Structure

Analog In (ADC) JSON

```
{
  "dev_type": 3,
  "dev_name": "Moter feedback",
  "id_room":"",
  "id_item": 17,
  "gpio": 5,
  "resln_bit": 10
}
```

ii. Key Value description

S.N.	Key	Details	Type	Data
1	gpio	ADC Input GPIO Pin	Number	5
2	resolution	ADC bit resolution	Number	8,10,12

d. PWM

i. JSON Structure

PWM JSON

```
{
        "dev_type": 5,
        "dev_name": "Dimmer",
        "id_room":"",
        "id_item": 21,
        "gpio": 3,
        "pwm_resln": 8,
        "freq_hz": 50,
        "duty_cycle": 0
}
```

ii. Key value description

S. N.	Key	Details	Type	Data
1	gpio	PWM output pin	Number	Any valid and PWM capable gpio pins
2	pwm_resln	Set the resolution of the ADC	Number	Default value to 8-bit resolution and can go up to 14-bit for ESP32S3 and ESP32C3, and up to 20-bits for ESP32
3	freq_hz	Setting PWM frequency in hertz	Number	50 - 10000
4	duty_cycle	Setting the initial duty cycle of the PWM output	Number	Defaults to 0 and can increase up to 100% i.e. 0 - 100

e. UART

i. JSON Structure

UART JSON

```
{
        "dev_type": 6,
        "dev_name": "MB1013 ultrasonic sensor",
        "id_room": 35,
        "id_item": 22,
        "gpio_tx": 17,
        "gpio_rx": 18,
        "baud": 9600
}
```

ii. Key value description

S. N.	Key	Details	Type	Data
1	gpio_tx	UART TX enabled pin	Number	17 or another UARt tx pins
2	gpio_rx	UART RX enabled pin	Number	18 or another UARt rx pins
3	baud	Transmission and receiving baud rate for UART communication	Number	600 , 1200 , 1800 , 2400 , 4800 , 9600, 19200 , 28800, 38400, 57600, 76800, 115200, 230400, 460800, 576000, 921600

f. One Wire

i. JSON Structure

OneWire JSON

```
{
  "dev_type": 7,
  "dev_name": "Humidity Kitchen",
  "id_room":"",
  "id_item": 15,
  "gpio": 17
}
```

ii. Key value description

S.N.	Key	Details	Type	Data
1	gpio	GPIO pin number for One-Wire Data	Number	17

g. I2C

i. JSON Structure

I2C JSON

```
{
  "dev_type": 8,
  "dev_name": "Sensor for my Dog",
  "id_room":"",
  "id_item": 5,
  "gpio_sda": 21,
  "gpio_scl": 22,
  "pullup_scl": true,
  "pullup_sda": true,
  "slave_addr": 105
}
```

ii. Key Value description

S.N.	Key	Details	Type	Data
1	gpio_sda	I2C SDA pin GPIO number	Number	21
2	gpio_scl	I2C SCL pin GPIO number	Number	22
3	pullup_sda	Enable/Disable internal Pull-up on SDA Pin	Boolean	True : Enable, False : Disable
4	pullup_scl	Enable/Disable internal Pull-up on SCL Pin	Boolean	True : Enable, False : Disable
5	slave_addr	Device Slave address	Number	105

h. SPI

i. JSON Structure

SPI JSON

```
{
  "dev_type": 9,
  "dev_name": "BMP280 Living room",
  "id_room":"",
  "id_item": 13,
  "gpio_miso": 12,
  "gpio_mosi": 13,
  "gpio_sck": 14,
  "gpio_cs": 15
}
```

ii. Key Value description

S.N.	Key	Details	Type	Data
1	gpio_miso	SPI MISO pin GPIO number	Number	12
2	gpio_mosi	SPI MOSI pin GPIO number	Number	13
3	gpio_sck	SPI Clock (SCK) pin GPIO number	Number	14
4	gpio_cs	SPI slave select pin GPIO number	Number	15

i. Other

i. JSON Structure

Device type Other JSON

```
{
        "dev_type": 10,
        "dev_name": "Main gate sensor",
        "id_room":"",
        "id_item": 18,
        "en_gpio1": true,
        "gpio1": 3,
        "en_gpio2": false,
        "gpio2": 0
        "en_gpio3": false,
        "gpio3": 0
}
```

ii. Key Value description

S.N.	Key	Details	Type	Data
1	en_gpio1	Enable or disable the GPIO	Boolean	True : GPIO1 enabled, False : GPIO1 not enabled
2	gpio1	GPIO Input pin number	Number	3

3	en_gpio2	Enable or disable the GPIO	Boolean	True : GPIO2 enabled, False : GPIO2 not enabled
4	gpio2	GPIO Input pin number	Number	0
5	en_gpio3	Enable or disable the GPIO	Boolean	True : GPIO3 enabled, False : GPIO3 not enabled
6	gpio3	GPIO Input pin number	Number	15

4. ## EzloPi Partition Details

Ezlopi Partition is designed for standard 4MB modules. The details regarding address for different variant of chips are in the table below.

S. N.	Binary Details	Partition Type	ESP32 (4MB)	ESP32-S3 (4MB)	ESP32-S3 (8MB)	ESP32-C3 (4MB)	Size	Publised file names
1	Bootloader	Default system partition, non alterable address	0x1000	0x000	0x000	0x000	N/A	0x1000.bin or ox000.bin
2	OTA Initial Data	Default system partition	0x15000	0x15000	0x15000	0x15000	4KB	0x15000.bin
3	Partition Table Data	Default system partition, non alterable address	0x8000	0x8000	0x8000	0x8000	N/A	0x8000.bin
4	EzloPi App Binary	Custom partition for EzloPi	0x20000	0x20000	0x20000	0x20000	1983KB	0x20000.bin
5	Provisioning Data	Custom partition for EzloPi	0x18000	0x18000	0x18000	0x18000	32KB	Generated on cloud call

5. ## Flash memory map for provisioning data

Ezlopi has different firmware binaries to be flashed into the ESP32 based devices as per partition planned. Among all these application binaries we use one binary for storage of provisioning data. The binary is commonly named as prov_data.bin and whose flash address is 0x18000 and memory map is listed below :

Info group	Info Name	Data type	Offset (Hex)	Size (Dec)	Size (HEX)	Mandatory	Description
Connection-Data **Offset: 0x0000** **Size: 0x6000**	S.N.	int16	0x00	2	0x02	-	Serial Number
	Config Version	int16	0x02	2	0x02	+	Config Version for provisioning Data
	Provisioning UUID	String	0x04	126	0x7E	+	
	UUID	String	0x82	126	0x7E	+	
	Provisioning Server	String	0x100	256	0x100	+	
	Provisioning token	String	0x200	512	0x200	+	
	Cloud Server	String	0x400	256	0x100	+	
	Reserve	-	0x500	2816	0xB00	-	
	EzloPi Config	String	0x1000	4096	0x1000	+	EzloPi Config JSON
	Reserve	-	0x2000	4096	0x1000	-	
	CA Certificate	String	0x3000	4096	0x1000	+	
	SSL Private Key	String	0x4000	4096	0x1000	+	
	SSL Shared Key	String	0x5000	4096	0x1000	+	
HUB-Data **Offset: 0x6000** **Size: 0x2000**	S.N.	int16	0x00	2	0x02	-	
	Version	int32	0x02	4	0x04	-	EzloPi Firmware Version
	Build	int16	0x06	2	0x02	-	EzloPi Firmware Build
	ID	int64	0x08	8	0x08	+	
	Reserve	-	0x10	8	0x08	-	
	ZWave Region	String	0x18	8	0x08	-	
	SSID	String	0x20	32	0x20	+	
	Password	String	0x40	64	0x40	+	
	Device Mac	String	0x80	64	0x40	+	
	Name	String	0xC0	64	0x40	+	
	Manufacturer	String	0x100	64	0x40	-	
	Brand	String	0x140	64	0x40	-	
	Model	String	0x180	64	0x40	-	
	EzloPi Device Type	String	0x1C0	64	0x40	-	
	Reserve	-	0x200	7680	0x1E00	-	